A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously revisits neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is thought to support systems consolidation, the process by which memory retrieval becomes progressively more dependent on neocortical and less dependent on hippocampal circuits. Evidence indicates that reactivation is not indiscriminate but preferentially targets particular memory traces.

Systems consolidation has motivated an extensive modeling literature addressing spontaneous reactivation, replay prioritization, and the hippocampo-neocortical redistribution of memory representations. In parallel, a growing literature has described hippocampal and neocortical circuits within the predictive-coding (PC) framework. Here, we connect these two lines of work by showing how coupled PC networks can generate autonomous intrinsic activity that redistributes stored memory content between neural systems.

We model the hippocampo-neocortical system as two coupled PC associative memory networks. The asymmetric coupling is inspired by the interaction modalities observed between hippocampal area CA1 and the prefrontal cortex, combining bottom-up recruitment with effective top-down suppression. This coupling establishes a closed-loop dialogue in which hippocampal activity trains the neocortical network, while neocortical feedback biases subsequent hippocampal activity toward content that remains poorly represented.

Under simplifying assumptions and after neural activity has settled, the coupled dynamics identify the direction in the hippocampal memory manifold with the largest remaining neocortical reconstruction error. A subsequent local neocortical plasticity event changes the synaptic weights so as to produce the largest immediate reduction in this error. Repeated reactivation and plasticity therefore greedily minimize the largest representational gap between the two systems. As a direction becomes represented neocortically, the discrepancy that prioritized it declines, allowing reactivation to shift toward less well-represented content.

In simulations with correlated memories, the architecture transferred memory content faster than random memory rehearsal. Together, these results provide a PC account on how the neocortical influence could guides hippocampal reactivation, prioritizing content that remains weakly represented.

Author summary

1 Introduction

check for tense coherence

An important fraction of the brain's metabolic budget is devoted to intrinsic activity, neuronal dynamics generated spontaneously by the brain in the absence of specific sensory input or behavioural demands [1, 2]. One of the best characterised events taking place during intrinsic activity is neural reactivation: the reinstantiation, during sleep or quiet rest, of activation patterns observed during waking experience (REF). For a substantial fraction of these events, activity in the hippocampus and neocortex is coordinated, giving rise to what is termed the hippocampo-cortical dialogue [3, 4].

This dialogue supports systems consolidation, the process by which memory retrieval becomes progressively less dependent on the hippocampus and more dependent on neocortical circuits [?, ?, 3, 5]. Systems consolidation is therefore often described as a form of memory transfer, whereby the neocortex progressively acquires memory content initially encoded in the hippocampus (REF).

Extensive literature indicates that the reactivation events underlying systems consolidation preferentially target certain memory traces over others. Traces that are recent, that correspond to novel experiences, or that **remain weakly encoded in cortex are prioritized for reactivation (REF)**. The precise neural mechanisms inducing this prioritization remain poorly understood (REF).

Reactivation and systems consolidation have motivated a broad range of computational models, from complementary learning systems [?] to autonomously reactivating attractor networks [?, ?, ?], generative replay [?, ?], and utility, context, or familiarity based replay scheduling [?, ?, ?, ?, ?]. These accounts differ in what drives reactivation, such as trace strength, recency, attractor competition, context, reward, or expected utility, and what information is transferred. Here, we focus on a single candidate mechanism, grounded in a specific communication motif between the hippocampus and neocortex.

The communication motif we consider has been most clearly characterized between the prefrontal cortex (PFC) and hippocampal area CA1, and combines two directional interactions: a bottom-up drive from CA1 to PFC and a top-down suppression from PFC to CA1. **During NREM sleep, a brief synchronized burst within a CA1 neuronal assembly is associated with increased firing in functionally coupled PFC populations (REF). Conversely, when PFC populations reactivate independently, without a coincident CA1 burst, CA1 neurons that normally participate in coordinated CA1–PFC events are preferentially suppressed (REF). Together, these findings motivate an asymmetric, representation-selective dialogue: hippocampal reactivation recruits selected prefrontal populations, whereas independent prefrontal reactivation suppresses the corresponding hippocampal representation. This motif may extend beyond PFC–CA1: entorhinal projections also recruit local inhibitory circuits in CA1, providing an potential additional pathway for similar bi-directional interation (REF).**(change place maybe)

Wang, Poe, and Zochowski gave this motif a functional interpretation: cortical memory representations regulate hippocampal reactivation according to their degree of consolidation [6]. In their model, as a cortical representation forms during reactivation, it recruits hippocampal inhibitory interneurons that selectively suppress the corresponding hippocampal trace. As consolidation proceeds, each trace withdraws its own reactivation, leaving replay to the content that still lacks cortical support.

We extend this account in three ways. First, we develop the functional role of feedback inhibition in organizing memory redistribution. Second, we formulate the interacting hippocampal and neocortical memories within the predictive coding (PC) framework. Third, we show that, after both network states have converged, the resulting local plasticity rule takes a gradient step on a candidate objective for memory redistribution.

In the proposed model, neocortical feedback inhibition biases reactivation away from hippocampal content already supported cortically and toward content that remains poorly represented. This dynamic directs the consolidation process toward the largest representational gap between the two systems. We show that, within the PC framework and under simplifying assumptions, one neocortical plasticity event evaluated after activity convergence follows the negative gradient of the largest settled reconstruction error over the hippocampal memory manifold. The resulting architecture gives rise to prioritized reactivation, redistributing memory content as consolidation proceeds, and terminating reactivation once redistribution is complete.

More broadly, this work contributes to the study of fast and reliable offline information transfer between artificial neural networks.

We deliberately adopt a simplified model in which the hippocampus and neocortex are each reduced to a single associative memory network of continuous activity units rather than spiking neurons; the resulting analytical tractability is what allows the mechanism to be derived rather than assumed. The analysis is performed on a linear version of the network, and the results are then generalized numerically for nonlinear activation functions.

2 Models

In this section, we introduce the network architecture, beginning with its local dynamics and their coupling. We then present the network training and memory consolidation dynamics.

The model consists of two coupled associative-memory PC networks (PCNs), the teacher and student networks, corresponding respectively to the hippocampus and neocortex. Taken independently, each network performs retrieval by carrying out inference on a given cue, minimizing its local free energy through changes in neuronal activity. Each network also learns by minimizing its local free energy, this time through changes in its synaptic weights (REF). The full architecture adds an asymmetric coupling between the teacher and student networks.

Memory storage occurs in two successive phases. The teacher first acquires externally imposed activity patterns; the coupled network then transfers this content to the student during intrinsic consolidation (Sec. 2.3).

2.1 A single predictive-coding associative memory

Tang et al. introduced the implicit covariance-learning PCN (implicit covPCN), which constitutes the basic building block of our architecture [7]. The implicit covPCN was developed as a more biologically plausible alternative to the explicit covPCN, whose learning rule requires non-local matrix operations and becomes numerically unstable on structured data [?, ?, ?, 7]. Here we present the linear version of the network which is the basis of the analytical study presented in this work.

Both the teacher T and the student S, corresponding respectively to the hippocampus and neocortex, are implicit covPCNs with dense recurrent connectivity and no self-connections. Each performs associative memory through local inference and plasticity.

We use $\alpha \in \{S, T\}$ to index either network. For activity $x_\alpha \in \mathbb{R}^d$ and recurrent weights W_α, define the prediction-error operator and error

$$M_\alpha = I - W_\alpha, \qquad \varepsilon_\alpha = M_\alpha x_\alpha, \tag{1}$$

and the free energy the network minimizes,

$$F_\alpha^{\text{rec}}(x_\alpha; W_\alpha) = \frac{\pi_\alpha}{2}\|\varepsilon_\alpha\|^2 = \frac{\pi_\alpha}{2}\|M_\alpha x_\alpha\|^2, \tag{2}$$

where $\pi_\alpha > 0$ is the precision of the prediction. We call F_α^{rec} the recurrent free energy because it accounts only for the network's internal prediction error generated by its recurrent connectivity. Network α lowers F_α^{rec} along two routes on two timescales: fast *inference* changes its activity, while slow *plasticity* changes its weights,

$$\tau_\alpha \, \dot{x}_\alpha = -\pi_\alpha \, M_\alpha^\top \varepsilon_\alpha, \qquad \dot{W}_\alpha = \eta \, \pi_\alpha \, \varepsilon_\alpha \, x_\alpha^\top, \tag{3}$$

with activity time constant τ_α and learning rate η.

The prediction may optionally pass through a pointwise nonlinearity f, giving $\varepsilon_\alpha = x_\alpha - W_\alpha f(x_\alpha)$. We take the linear case $f = \mathrm{id}$ throughout the analysis and return to the rectified case $f = \mathrm{ReLU}$ in Section ??. In addition, as introduced in the next section, the teacher activity is normalized such that $|x_T| = 1$ after each integration step.

Each network is trained according to the plasticity dynamics in Eq. (3), either during interleaved clamping or through intrinsic activity during system consolidation (2.3). In a fully trained network α, the stored content is the kernel of its error operator,

$$\mathcal{U}_\alpha = \ker M_\alpha = \{x \in \mathbb{R}^d : \, M_\alpha x = 0\}, \tag{4}$$

the network's memory manifold. Every stored pattern satisfies $M_\alpha x = 0$ and has zero recurrent free energy F_α^{rec} [7].

For the linear network, $\mathcal{U}_\alpha$ is a linear subspace. Any linear combination of stored memories is therefore a zero-energy fixed point. The network stores a flat memory manifold rather than a set of point attractors as in classical Hopfield networks (REF).

2.2 Coupling the teacher and the student

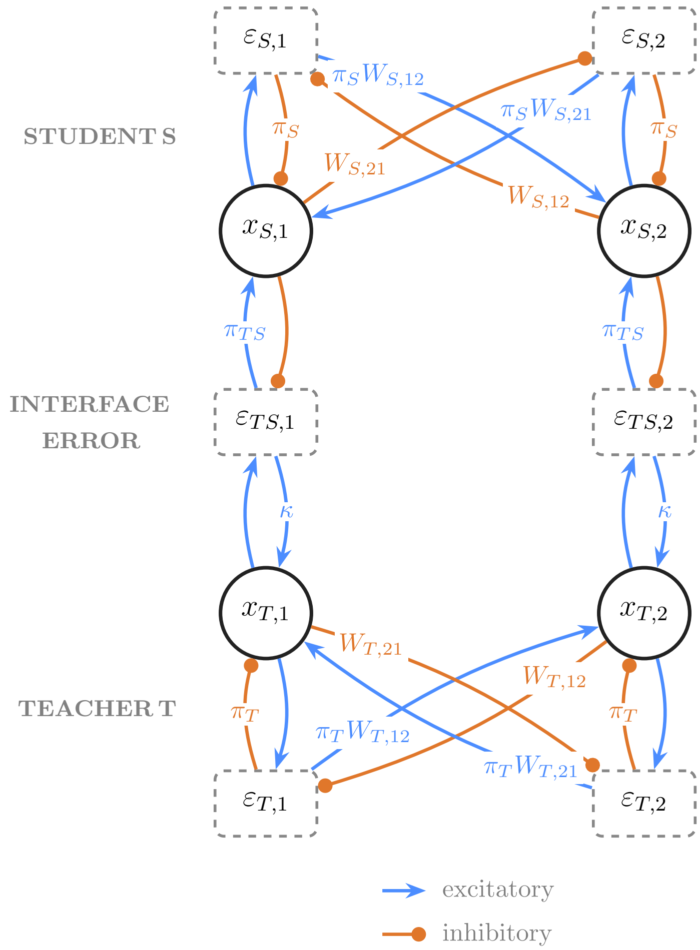

Fig 1. Model architecture (two units per population). Student S (top, cortical) above teacher T (bottom, hippocampal): value units x, recurrent errors $\varepsilon_S, \varepsilon_T$, and the one-to-one interface error $\varepsilon_{TS} = x_T - x_S$. Connections without labels have scalar value 1. During replay, the teacher-side interface gain κ makes the teacher ascend the student deficit rather than descending it as in classical PC networks.

Figure 1 shows the full architecture formed by coupling the student S and teacher T.
The two networks interact through a one-to-one *interface error*,

$$\varepsilon_{TS} = x_T - x_S, \tag{5}$$

which measures the mismatch between the student state and the teacher state currently expressed. Without the interface influence, both the teacher T and the student S activities follow the inference dynamics in Eq. (3) and descends their own recurrent free energy F_T^{rec}.

For the teacher, the coupling adds a direct interface-error drive, giving

$$\tau_T \dot{x}_T = \underbrace{-\pi_T M_T^\top \varepsilon_T}_{-\nabla_{x_T} F_T^{\text{rec}}} + \kappa \varepsilon_{TS}, \qquad \|x_T\| = 1, \tag{6}$$

where κ is the teacher-side coupling gain. During intrinsic activity, $\kappa > 0$, so the interface-error population excites the teacher. In conventional predictive coding, minimizing the interface error with respect to x_T would instead contribute $-\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. After each integration step, teacher activity is normalized to unit norm to prevent the linear dynamics from producing unbounded activity. This operation idealizes fast population gain control: in biological circuits, divisive normalization mediated by recurrent and feedforward inhibition constrains the overall response magnitude while largely preserving the relative activity pattern that carries the represented content [**?**].

The student side retains the conventional PC dynamics. Student activity minimizes its recurrent free energy and receives an excitatory correction from the interface-error population,

$$\tau_S \dot{x}_S = -\pi_S M_S^\top \varepsilon_S + \pi_{TS}\varepsilon_{TS}, \tag{7}$$

the student activity is not normalized during the simulation of the whole model.

Both contributions to Eq. (7) are gradient-descent terms on the student free energy functional which can therefore be expressed as

$$F_S(x_S, x_T; W_S) = \underbrace{\tfrac{\pi_S}{2}\|M_S x_S\|^2}_{F_S^{\mathrm{rec}}} + \underbrace{\tfrac{\pi_{TS}}{2}\|x_T - x_S\|^2}_{F_{TS}}, \qquad \tau_S \dot{x}_S = -\nabla_{x_S} F_S, \tag{8}$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. The student weights follow the local plasticity rule

$$\dot{W}_S = \eta \pi_S \varepsilon_S x_S^\top = -\eta \nabla_{W_S} F_S. \tag{9}$$

Activity and plasticity evolve concurrently in the simulations.

The overall influence between S and T deserve carefull reading. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T - x_S$ with a negative sign, inhibiting the interface error. The combined pathway $x_S \to \varepsilon_{TS} \to x_T$ is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional PC network, the interface error would instead inhibit the lower population., producing top down excitation through disinhibition.

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2.3 Initial storage and consolidation dynamics

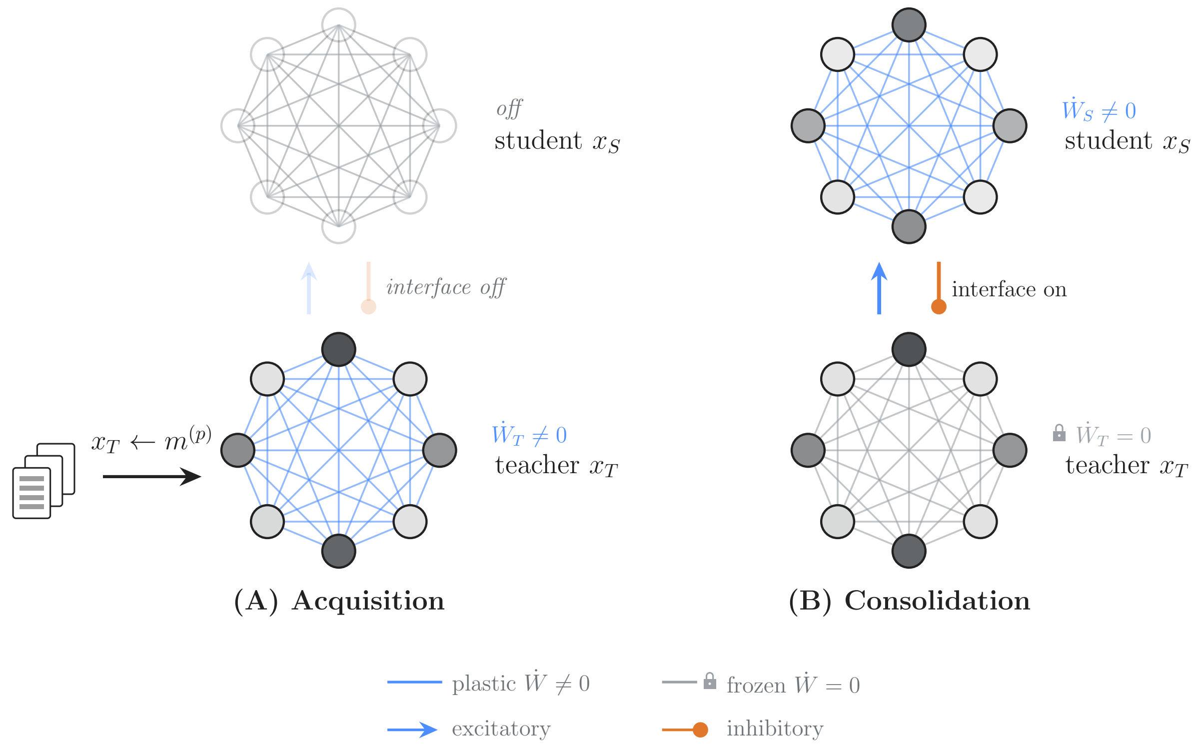

Fig 2. Initial storage and consolidation dynamics. Initial storage creates the teacher memory manifold from externally imposed patterns. Subsequent intrinsic dynamics use this manifold to generate student training states while preserving the teacher representation. The student is therefore trained by the coupled network dynamics rather than by direct clamping.

Memory is stored in the two networks by the same local recurrent plasticity rule (Eq. 3), but under two different regimes for generating neuronal activity. As shown in Fig. 2, the first phase establishes the teacher memory; the second redistribute (or transfer) this content to the student without presenting the student with an external target.

During initial storage (Fig. 2(A)), teacher activity x_T is clamped to memory patterns $m^{(p)}$ in interleaved order while W_T follows Eq. 3. The student and the interface remain inactive. These updates reduce the teacher's recurrent prediction error on the presented patterns and establish its memory manifold $\mathcal{U}_T = \ker M_T$ (Methods). Interleaved training allows correlated memories to be stored while mitigating interference (REF). In contrast to our model, the hippocampus does not appear to require interleaved training to reliably store memory content during awake activity (REF). The orthogonalization of memory representations, particularly in the granule-cell layer of the dentate gyrus, is thought to support rapid learning of correlated memory traces (REF). This biological machinery is not part of our model and is therefore replaced by interleaved training.

During consolidation (Fig. 2(B)), the external clamp is removed and the interface is activated. The teacher and student activities follow Eqs. (6) and (7), while W_T remains fixed and only W_S follows the plasticity rule in Eq. (9). The student is never clamped: teacher-driven intrinsic activity supplies the states stored by its local recurrent plasticity and influence the student through the interface error.

3 Results

3.1 Reactivation and memory content redistribution

We first examined whether intrinsic activity in our model could reactivate and redistribute stored memory content.

We stored three unit-normalized MNIST patterns in a linear 784-unit teacher network, froze the teacher's recurrent weights, and started the coupled dynamics with a plastic student whose recurrent weights were initialized to zero. During this intrinsic phase, neither network received external input. Low-amplitude Gaussian noise was applied to the teacher activity, allowing the dynamics to explore the teacher memory manifold. The results are shown in Fig. 3.

Fig. 3(A, bottom), (B, top), and (C) show that teacher activity is dominated by mixture states rather than by reactivation of individual patterns. This behavior follows from the linear activation function: the stored patterns span a flat memory manifold containing every signed linear combination of those patterns, including their antipodal states.

Fig. 3(D) and (E) provide direct evidence of memory-content redistribution during intrinsic activity. The successive reduction of the three eigenvalues indicates that the student acquires the flat teacher memory manifold stages by stages across independent directions, rather than through a uniform reduction of error accross all directions. redistribution rate slows once most of the manifold is represented by the student and remains incomplete over the displayed interval. Longer simulations further reduce the residual redistribution gap, although at a much lower rate. At this late stage, the mismatch-driven selection signal is weak and the teacher increasingly relies on Gaussian-noise exploration to sample the remaining poorly represented directions. Despite this incomplete redistribution, the isolated-student probes in Fig. 4 show that, after our consolidation protocol, the student succesfully store the original patterns. Memory reconstruction capacities of the student are uneven at the partial checkpoint but allow recognizable reconstruction of all three memories later during consolidation.

Fig. 3(B, bottom) and (E, middle and bottom) show how the signals driving learning and reactivation attenuate during memory redistribution. The student plasticity rate decreases as the student–teacher mismatch contracts, revealing a self-limiting learning process: as the student acquires teacher-supported content, it removes the errors that generate further weight updates. In parallel, the teacher-side selection drive, generated by the mismatch-dependent top-down influence of the student on the teacher, also decreases. Three transient peaks accompany the successive reduction of the three error modes in Fig. 3(D), consistent with the sequential acquisition of the three independent directions spanning the rank-three teacher memory manifold. Both signals remain nonzero at the end of the displayed interval, so this run approaches but does not reach exact self-extinction.

To test whether this mechanism extends beyond redistribution from a fixed teacher memory set, we allowed the teacher repertoire to grow during an otherwise continuous consolidation run (Fig. 5). Each newly added memory introduced a new high-eigenvalue error mode, while the eigenvalues associated with memory directions already redistributed to the student remained close to zero. These new gaps then declined without resetting the student or discarding its previously learned weights, showing that the same mismatch-dependent dynamics can repeatedly redirect reactivation toward newly unsupported content. This protocol captures a feature of systems consolidation: ongoing experience continually adds new hippocampal memory content, which must then be integrated into neocortical circuits during subsequent periods of intrinsic activity (REF). In the model, each newly stored hippocampal direction automatically recreates a hippocampo-cortical mismatch. Renewed prioritization therefore requires neither an explicit replay queue nor an externally assigned novelty tag. As cortical support for the new content develops, both the mismatch and the associated learning drive recede.

In summary, the coupled architecture redistributes memory content from the teacher to the student during intrinsic activity. redistribution remains incomplete in the displayed run, yet the isolated-student probes show recognizable completion of all three stored patterns at the late checkpoint. Exact equality between the teacher and student memory manifolds is therefore not required for successful recall in this simple task. This result should nevertheless be interpreted as evidence of functional recall in a small associative-memory setup, rather than as a general demonstration of functional equivalence between the student and the teacher networks for more complex tasks. Tests involving classification of the reconstructed patterns, weaker cues, or more complex memory sets will be needed to determine when the residual redistribution deficit becomes functionally limiting. In the Discussion, we consider possible mechanisms for improving redistribution exhaustiveness.

The observed dynamics are consistent with neocortex-dependent prioritization. The mismatch signal biases teacher activity toward teacher-supported directions that remain poorly represented by the student, and this bias weakens as those directions are learned. This provides a possible mechanistic account of the preferential reactivation of recently encoded hippocampal representations during post-learning sleep (REF), followed by reduced hippocampal involvement as memory retrieval becomes increasingly supported by neocortical circuits (REF). Nevertheless, the model does not treat recency itself as the priority signal. It predicts that reactivation depends more directly on the mismatch between hippocampal support and cortical representation.

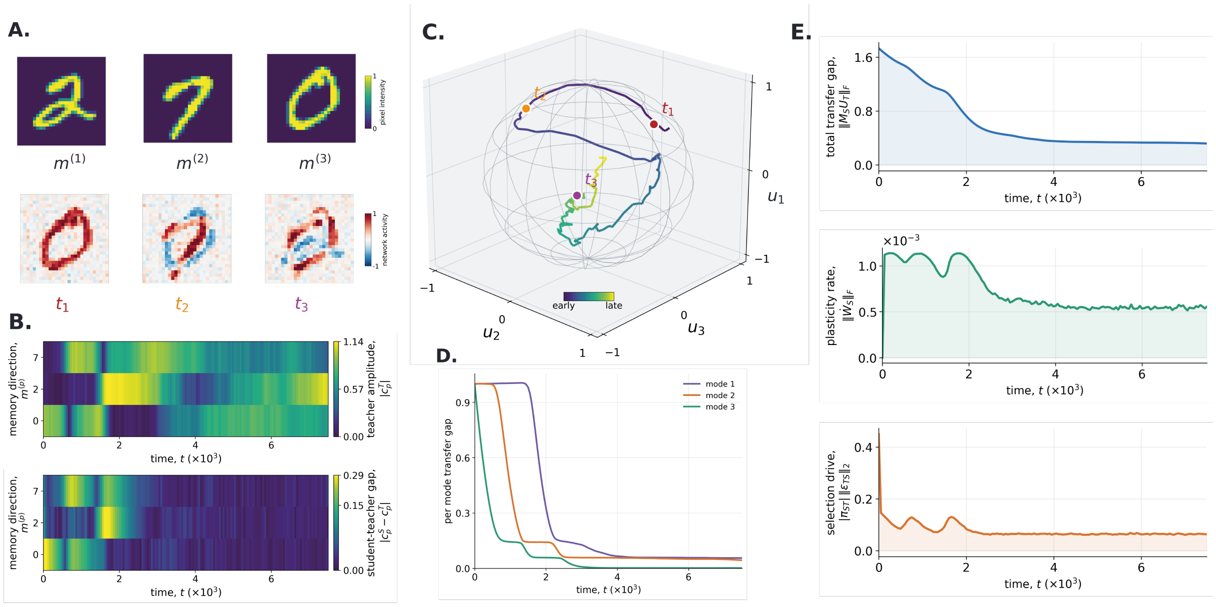

Fig 3. Spontaneous intrinsic activity progressively redistributes teacher-supported memory content. The frozen teacher network was coupled to a student network initialized with zero activity and zero recurrent weights. No external input was presented during the run; low-amplitude Gaussian noise seeded exploration of the teacher memory manifold. **A,** The three stored MNIST patterns (top; display order 2, 7, and 0) and teacher states at $t_1 = 250$, $t_2 = 1000$, and $t_3 = 7000$ (bottom). Although the stored patterns are positively signed, the linear network can express signed mixtures and antipodal states. **B,** (Top) Memory coefficient magnitude $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Method). Low values mean the teacher doesn't reactivates the memory while large values indicate strong reactivation. (Bottom) Student–teacher magnitude gaps $|c_p^S(t) - c_p^T(t)|$; a large gap indicates that the two networks express different magnitude along that pattern coordinate. **C,** Teacher activity projected onto an orthonormal basis U_T of the three-dimensional teacher memory space. The wireframe marks the unit sphere imposed by teacher normalization, and the colored markers identify the snapshots in panel A. Basis axes are orthogonal memory-space directions, not individual digits. **D,** Sorted eigenvalues of $G_S(t) = U_T^\top M_S(t)^\top M_S(t) U_T$. High eigenvalues identify teacher-supported directions poorly represented by the student, whereas low values indicate accurate storage of these components. **E,** (Top) Total redistribution gap $\|M_S U_T\|_F$, which measures how much of the teacher memory space remains unsupported by the student. (Middle) Student plasticity rate $\|\dot{W}_S\|_F$, the magnitude of the current recurrent-weight update. (Bottom) Teacher-side selection drive $|\pi_{ST}|\|\varepsilon_{TS}\|_2$, the magnitude of the mismatch-dependent signal acting on the teacher.

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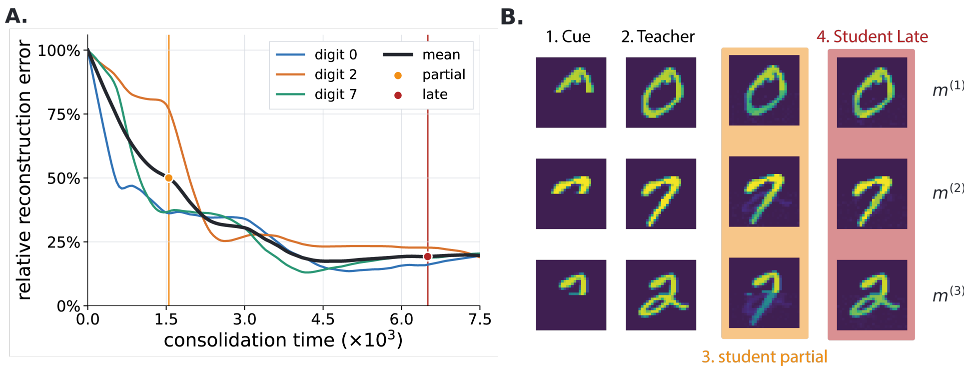

Fig 4. Transferred teacher memories become reconstructable by the isolated student. A, Relative recurrent reconstruction error for each stored MNIST pattern during consolidation. At each recorded time, the student weights were frozen, its state was set to the complete pattern $m^{(p)}$, and the error was evaluated as $r_p(t) = \|m^{(p)} - W_S(t)m^{(p)}\|_2/\|m^{(p)}\|_2$. Thus, 100% corresponds to the initially empty student with $W_S = 0$, whereas zero denotes exact recurrent prediction of that pattern. The black curve is the mean over the three digits; orange and red markers identify the partial ($t = 1550$) and late ($t = 6500$) checkpoints used in panel B. **B,** Clamped recall by isolated, frozen networks. From left to right, each row shows the partial cue, the teacher reconstruction, and student reconstructions at the partial and late checkpoints. The upper half of each digit was clamped to the stored pixels, the hidden half was initialized to zero, and the free units settled by minimizing the network self-energy $\frac{\pi}{2}\|(I - W_\alpha)x_\alpha\|_2^2$, with no teacher–student coupling or ongoing plasticity. Student recall performances are uneven at the partial checkpoint, in agreement with the reconstruction error in panel A. For the late checkpoint, all reconstruction outputs visually approach the teacher completions despite a remaining relative reconstruction error.

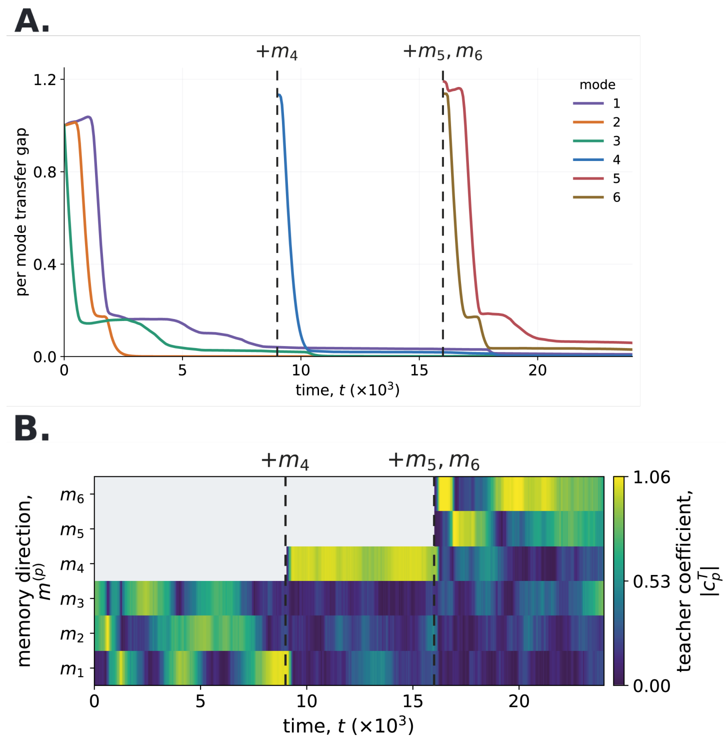

Fig 5. Staged memory acquisition renews reactivation and memory redistribution. A 50-unit linear teacher–student network was followed through one continuous run with six unit-normalized random patterns. The teacher initially stored m_1, m_2, and m_3; it acquired m_4 at $t = 9000$ and acquired m_5 and m_6 together at $t = 16000$ through interleaved teacher training on the whole corpus. Coupled intrinsic dynamics resumed after each acquisition without resetting student activity or recurrent weights. Dashed lines mark the acquisition events. **A,** Eigenvalues of the student error operator restricted to the currently supported teacher memory space. Previously transferred modes keep a small error, whereas each newly supported direction introduces a large transfer gap that subsequently decreases. **B,** Memory coefficient magnitudes $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Methods). Low values mean that the teacher does not reactivate the corresponding memory, whereas high values indicate strong reactivation. Newly added memories initially receive a stronger reactivation drive. As redistribution progresses, this drive decreases.

3.2 Fast memory content transfer of structured data

Having established that the coupled network dynamics can redistribute memory spaces, we next asked how efficiently this redistribution proceeds when the corpus contains unequally sized groups of correlated memories. We compared two protocols:

- **Intrinsic transfer (our method).** A trained, frozen teacher storing the memory space was coupled to an initially empty student. Teacher and student activity evolved under the coupled intrinsic dynamics, and the student updated its recurrent weights after each integration step.
- **Stochastic rehearsal.** At each plasticity event (time step), one memory was sampled uniformly from the indexed corpus and clamped directly onto the student before the same local recurrent update was applied.

Stochastic rehearsal emulates the redistribution dynamics occuring from non-prioritized reactivation of memories, comparing it with our method therefore allows us to assess the impact of prioritized reactivation on the redistribution dynamics.

The results for a correlated memory corpus are shown in Fig. 6.

We considered a dataset of 26 memories divided in three mutually orthogonal groups containing 18, 5, and 3 memories. Fig. 6(A) display the correlation structure of the memory set, memories within a group had a correlation of $\rho = 0.70$. The corpus construction, matched parameters, and implementation details are given in Materials and methods, Section 6.5.

We followed redistribution at two levels. The named-memory residual used in the heatmaps asks whether the student can reconstruct each stored example. We also reported a memory-subspace reconstruction error, R_{sub}, which give direct insight on how well the student network actually store the wanted memory manifold. We also checked, the worst-direction reconstruction error, R_{worst}, which isolates the single direction that the student store least accurately. Formal definitions of all three measures are given in Materials and methods, Section 6.5.

The two protocols initially reduced the emory-subspace reconstruction error at similar rates, but intrinsic transfer soon produced a faster decline than stochastic rehearsal (Fig. 6(B)).

Fig. 6(C) shows that intrinsic transfer reduced reconstruction errors at similar rates across all three groups, despite their unequal sizes, whereby stochastic rehearsal favor learning for the large clusters.

The difference in transfer rates reflects how each protocol selects the activity that drives plasticity. Stochastic rehearsal samples named memories uniformly. Because memories within each group are strongly correlated, these updates repeatedly reinforce their shared component. This induces faster learning of over-represented groups. On the other hand, intrinsic transfer is not tied to named-memory frequency. As the student learns a teacher-supported direction, the mismatch along that direction contracts and its selection drive weakens, allowing activity to shift toward directions with larger remaining deficits. The result is consistent with deficit-dependent reactivation allocating plasticity according to reconstruction error of the subspace rather than the sample size of a given correlated memory-cluster.

Fig. 7 shows how within-group correlation affects the redistribution rate of both protocols while memory group sizes remain fixed. At low correlation, stochastic rehearsal was faster, whereas increasing correlation shifted the advantage toward intrinsic transfer.

Stochastic rehearsal can be more efficient when correlations are weak. Because the corpus acts as a lookup table, each plasticity event is applied directly to an exact memory. Intrinsic transfer instead updates the student while activity moves through the teacher memory space, and these intermediate states may produce partly redundant updates. Direct presentation therefore retains an advantage when uniform sampling already covers the memory subspace evenly, which happens when correlation fall.

Together, these results show that deficit-dependent intrinsic dynamics can redistribute highly correlated memory content with fewer plasticity events than stochastic rehearsal. This advantage grows with correlation, when uniform sampling increasingly produces overlapping updates. Such efficiency would matter for the brain as energy and opportunities for offline reactivation are limited (REF).

The next section presents a mathematical account of this qualitative behavior. We ask whether the coupled dynamics can be understood as optimizing an objective function derived from the PC formulation.

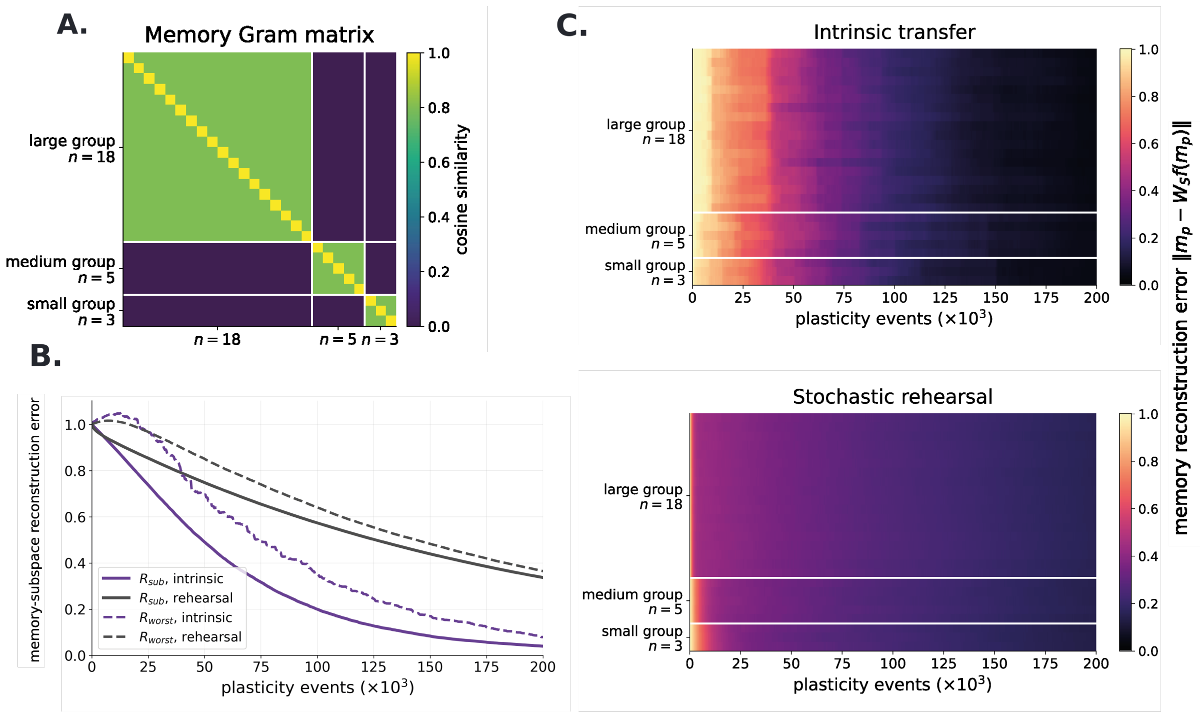

Fig 6. Intrinsic transfer accelerates redistribution across unequal groups of correlated memories. Both conditions used linear students initialized with $W_S = 0$, the same memory corpus, the same recurrent plasticity rule and rate, and 200,000 plasticity events. The corpus contained 26 memories divided into three mutually orthogonal groups of 18, 5, and 3 memories, with exact within-group cosine similarity $\rho = 0.80$. **A,** Memory Gram matrix. Each cell shows the cosine similarity between two memories. **B,** Reconstruction error over the complete memory space. Solid curves show the RMS memory-subspace error R_{sub}, which gives equal weight to every independent direction. Dashed curves show the worst-direction error R_{worst}, defined by the memory-space direction reconstructed least accurately by the student. Both measures equal one for the initially empty student, and lower values indicate more complete redistribution. After a short initial transient, intrinsic transfer reduced both errors faster than stochastic rehearsal and reached substantially lower final values. The rehearsal errors were still decreasing at the end of the protocol. **C,** Reconstruction errors for individual memories during intrinsic transfer (top) and stochastic rehearsal (bottom). Light colors indicate high reconstruction error and dark colors indicate accurate reconstruction. Intrinsic transfer reduced errors at similar rates across all three groups, whereas under stochastic rehearsal, errors remained higher for longer in the smaller groups. Metric definitions and full protocol details are provided in Materials and methods, Section 6.5.

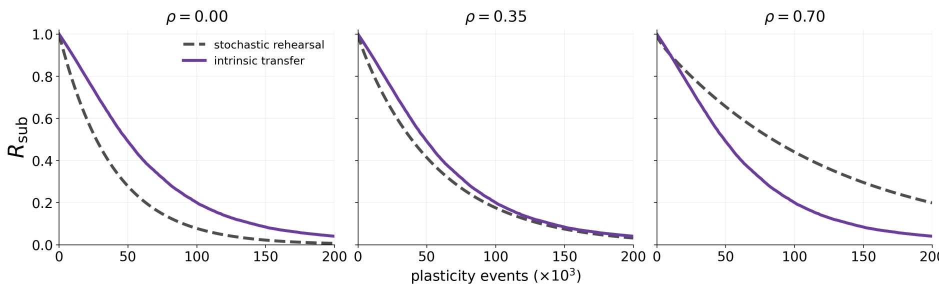

Fig 7. Increasing within-group correlation shifts the advantage from stochastic rehearsal to intrinsic transfer. The three conditions used the same group sizes (18, 5, and 3), student initialization, plasticity parameters, and number of plasticity events. Only the exact within-group cosine similarity changed, with $\rho \in \{0, 0.35, 0.70\}$. Each curve shows the RMS memory-subspace reconstruction error R_{sub}, with lower values indicating more complete redistribution. Gray dashed curves show stochastic rehearsal and purple curves show intrinsic transfer. At $\rho = 0$, stochastic rehearsal reduced the error faster. At $\rho = 0.35$, the two protocols performed similarly. At $\rho = 0.70$, intrinsic transfer became faster after a brief initial period and reached a substantially lower final error. As correlation increased, stochastic rehearsal slowed because uniformly sampled memories produced increasingly overlapping updates, whereas the intrinsic-transfer curves changed comparatively little. All conditions used linear units. Full definitions and protocol details are provided in Materials and methods, Section 6.5.

3.3 Approximating minimization of the largest settled neocortical free energy

The preceding results show that the coupled dynamics can redistribute memory content from the teacher to the student. We now ask whether redistribution progressively reduces the largest settled student free energy over the teacher memory manifold. We first show that, after student relaxation, the teacher-side interface drive is proportional to the gradient of the settled student free energy with respect to teacher activity. This result motivates two hypotheses:

1. **Activity selection:** the coupled dynamics approach a teacher state that nearly maximizes the settled student free energy over the teacher memory manifold.
2. **Synaptic plasticity:** a local student-plasticity event applied after activity relaxation approximates a gradient-descent step on this maximum.

Figure 8 tests these hypotheses numerically.

The following analysis and numerical tests consider the separation of timescales

$$\tau_S \ll \tau_T \ll \frac{1}{\eta}. \tag{10}$$

The first inequality allows student activity to relax while teacher activity changes little. This permits adiabatic elimination of the student activity, replacing it by its equilibrium for the current teacher state. The second inequality keeps synaptic weights approximately fixed during activity relaxation. The analytical derivation presented here uses the adiabatic limit, while the numerical tests assess this approximation at finite timescale separation.

As defined in Eq. (8), the student free energy F_S combines the student's recurrent prediction error with the mismatch between the teacher and student states. For fixed teacher activity x_T and student weights W_S, the linear student dynamics minimize F_S with respect to x_S. We denote the free energy remaining after this relaxation by

$$q(x_T; W_S) = \min_{x_S} F_S(x_S, x_T; W_S). \tag{11}$$

Under the adiabatic approximation, q therefore describes how the settled student free energy varies as teacher activity changes.

The relevant set of teacher states is the unit sphere of its memory manifold,

$$\mathbb{S}_T = \{x_T \in \mathcal{U}_T : \|x_T\| = 1\}, \qquad \mathcal{U}_T = \ker M_T. \tag{12}$$

The two parts of this constraint are implemented differently in the numerical tests. The teacher activity derivative is constrained to the tangent plane of the unit sphere, preserving unit norm. Membership in $\mathcal{U}_T$ is approximate: the teacher's recurrent dynamics damp activity outside its memory manifold, but the interface drive can produce departures from it. We therefore use $\mathbb{S}_T$ to define the optimization benchmark and measure distances using the full teacher activity vector. No noise is applied in these tests.

The largest settled student free energy over $\mathbb{S}_T$ is

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \tag{13}$$

By definition,

$$q(x_T; W_S) \le \mathcal{F}_{\max}(W_S; W_T) \qquad \text{for every } x_T \in \mathbb{S}_T.$$

Thus, $\mathcal{F}_{\max}$ bounds the settled student free energy over the entire teacher memory manifold. A low value means that the student settles to a low free energy for every state on this manifold, whereas a large value means that at least one state remains poorly represented.

Hypothesis **1** therefore predicts that, during intrinsic activity with fixed student weights W_S, the teacher approaches a state whose settled student free energy is close to this upper bound. We test this prediction by comparing the observed teacher state with an independently calculated ideal teacher state that attains the maximum.

This nested quantity also has an interpretation in terms of the complete teacher–student network free energy,

$$F_{\mathrm{net}}(x_T, x_S; W_T, W_S) = F_T^{\mathrm{rec}}(x_T; W_T) + F_S(x_S, x_T; W_S). \tag{14}$$

Since $\mathcal{U}_T = \ker M_T$, the teacher recurrent free energy vanishes for every $x_T \in \mathbb{S}_T$. Consequently,

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_{\mathrm{net}}(x_T, x_S; W_T, W_S) \tag{15}$$

The largest settled student free energy is therefore also the largest settled free energy of the complete network over this manifold.

We call a *redistribution cycle* the relaxation of teacher and student activity at fixed weights followed by one student-plasticity event. Hypothesis **2** predicts that, once the teacher has approached a state maximizing the settled student free energy and the student has settled in response to that state, the local update of the student weights approximates a gradient-descent step on $\mathcal{F}_{\max}$:

$$\boxed{\Delta W_S \approx -h P_0 \nabla_{W_S} \mathcal{F}_{\max}(W_S; W_T),} \qquad h = \eta\, \Delta t_{\mathrm{pl}}, \tag{16}$$

where Δt_{pl} is the event duration and P_0 sets the diagonal entries of a matrix to zero, corresponding to the absence of self-connections in our model.

This construction is close in purpose to classical predictive coding, while differing in how the states on which free energy is minimized are obtained. PC networks ordinarily minimize free energy for activity elicited by current sensory input; variation in those inputs supplies the states on which learning acts. During autonomous redistribution, no external input specifies the state to be learned. The network must therefore organize both exploration of the teacher memory manifold and local student learning. Exploration may temporarily increase the instantaneous network free energy, while successive student-plasticity events are expected to reduce the network settled free energy over the teacher memory manifold.

The settled interface error is proportional to the gradient with respect to teacher activity. The following result identifies how the student's equilibrium response determines the interface drive received by the teacher. Define

$$S_S = M_S^\top M_S, \qquad H_S = \pi_{TS} I + \pi_S S_S, \qquad N_S = I - \pi_{TS} H_S^{-1} = \pi_S S_S H_S^{-1}. \tag{17}$$

Theorem 1 (Settled student response and interface-driven ascent)**.** *For the linear student with* $\pi_S, \pi_{TS} > 0$*, fixed teacher activity, and fixed weights,* F_S *has the unique global minimizer*

$$x_S^*(x_T) = \pi_{TS} H_S^{-1} x_T = (I - N_S) x_T. \tag{18}$$

The settled student free energy and its gradient with respect to teacher activity are

$$q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T, \qquad \nabla_{x_T} q = \pi_{TS} N_S x_T = \pi_{TS}(x_T - x_S^*). \tag{19}$$

Thus, when the student is at equilibrium and $\kappa > 0$*, the teacher-side interface contribution is a positive multiple of this gradient:*

$$\kappa \varepsilon_{TS}^* = \frac{\kappa}{\pi_{TS}} \nabla_{x_T} q. \tag{20}$$

Proof. For fixed x_T, the student gradient and Hessian are

$$\nabla_{x_S} F_S = H_S x_S - \pi_{TS} x_T, \qquad \nabla_{x_S}^2 F_S = H_S \succ 0.$$

The positive-definite Hessian ensures that student gradient flow at fixed teacher activity converges to $x_S^*(x_T)$. Solving the stationarity equation gives Eq. (18). Expanding the energy and using $H_S x_S^* = \pi_{TS} x_T$ gives

$$q = \frac{\pi_{TS}}{2} x_T^\top x_T - \frac{\pi_{TS}}{2} x_T^\top x_S^* = \frac{\pi_{TS}}{2} x_T^\top N_S x_T.$$

Since N_S is symmetric, differentiating it gives $\nabla_{x_T} q = \pi_{TS} N_S x_T$. Finally, $x_T - x_S^* = N_S x_T$, which identifies the gradient with the settled interface error. □

The theorem shows that, once the student has settled, the teacher-side coupling favors changes in teacher activity that increase the settled student free energy. Within the teacher memory manifold, this favors states that remain poorly represented by the student. This result alone does not establish that the full coupled dynamics approach a teacher state whose settled student free energy is close to the maximum over $\mathbb{S}_T$. This is the prediction of Hypothesis **1**, which we test numerically below.

Coupled activity approaches states with near-maximal settled student free energy. We first tested whether the teacher approaches an *ideal teacher state*, defined by

$$x_T^* \in \arg\max_{x_T \in \mathbb{S}_T} q(x_T; W_S). \tag{21}$$

For the linear model, these states can be calculated directly from the weights, independently of the simulated trajectories, by finding the leading eigenspace of N_S restricted to the teacher memory space. The reference calculation is given in Materials and methods, Section 6.6.1.

In a two-dimensional memory space, teacher trajectories from different initial states approached the two ideal states, despite starting outside the memory manifold (Fig. 8A). We then tested selection across 20 independent networks with six-dimensional memory spaces and partially trained students. At each training checkpoint, student weights were frozen and the coupled activity relaxed from ten random teacher states. The mean final angle to the nearest ideal state remained small across the tested checkpoints, approximately 0.4° to 0.6° (Fig. 8B). These results support Hypothesis **1** at the finite timescale separation tested here: the coupled dynamics approach states close to those that maximize the settled student free energy.

Local student plasticity produces the predicted decrease in the maximum. We next tested the effect of one local plasticity event after activity relaxation. The observed student activity and recurrent error determine the update

$$\Delta W_S = h\pi_S P_0\big[\varepsilon_S x_S^\top\big], \qquad \varepsilon_S = M_S x_S. \tag{22}$$

If this update follows the negative gradient of $\mathcal{F}_{\max}$, its first-order effect is

$$\mathcal{F}_{\max}(W_S; W_T) - \mathcal{F}_{\max}(W_S + \Delta W_S; W_T) \approx \frac{\|\Delta W_S\|_F^2}{h}. \tag{23}$$

The squared update norm appears because a gradient step decreases a differentiable objective, to first order, by the step size times the squared gradient norm. At student equilibrium and an ideal teacher state, this prediction follows from differentiating the maximized settled energy, provided its leading eigenvalue is locally simple (Materials and methods, Section 6.6.3).

Across the frozen configurations used for the selection test, the measured decreases closely followed this prediction (Fig. 8C). The maximum was recomputed after every update, allowing the maximizing teacher state to change. Thus, the agreement concerns the largest settled free energy over the entire teacher memory manifold. It supports the predicted effect of local gradient descent in Hypothesis **2**, although agreement in this scalar measure alone does not establish exact alignment of the full weight update with the gradient.

Repeated redistribution cycles reduce the largest remaining deficit. Finally, we tested whether successive activity-selection and plasticity events reduce $\mathcal{F}_{\max}$. Both intrinsic transfer and random-state rehearsal started from the same student weights, constructed to store five of the six teacher memories. This left one independent memory direction incompletely represented. Random-state rehearsal sampled unit states uniformly from the teacher memory manifold and used the same local plasticity rule and step size as intrinsic transfer.

Intrinsic transfer reduced the maximum more rapidly over successive events (Fig. 8D). After 700 events, the median maximum settled free energy was approximately 20% of its initial value under intrinsic transfer, compared with approximately 86% under random-state rehearsal. Because the reference samples the memory sphere uniformly, this advantage does not depend on unequal sampling frequencies of named memories. It is consistent with directing updates toward the remaining deficit when much of the memory space is already represented. The comparison measures progress per plasticity event; it does not establish an advantage in elapsed relaxation time.

Complete redistribution would require $\mathcal{F}_{\max} = 0$, which is equivalent to the student representing the entire teacher memory manifold,

$$\mathcal{F}_{\max} = 0 \quad \Longleftrightarrow \quad \mathcal{U}_T \subseteq \ker M_S. \tag{24}$$

The residual maximum in Fig. 8D remains nonzero. The results therefore support approximate minimization through repeated local updates in this linear, noiseless, settle-then-update regime, without establishing convergence to complete redistribution. We now consider how this dependence of reactivation on the student's remaining deficit may contribute to biological memory consolidation, and where the assumptions of the model limit that interpretation.

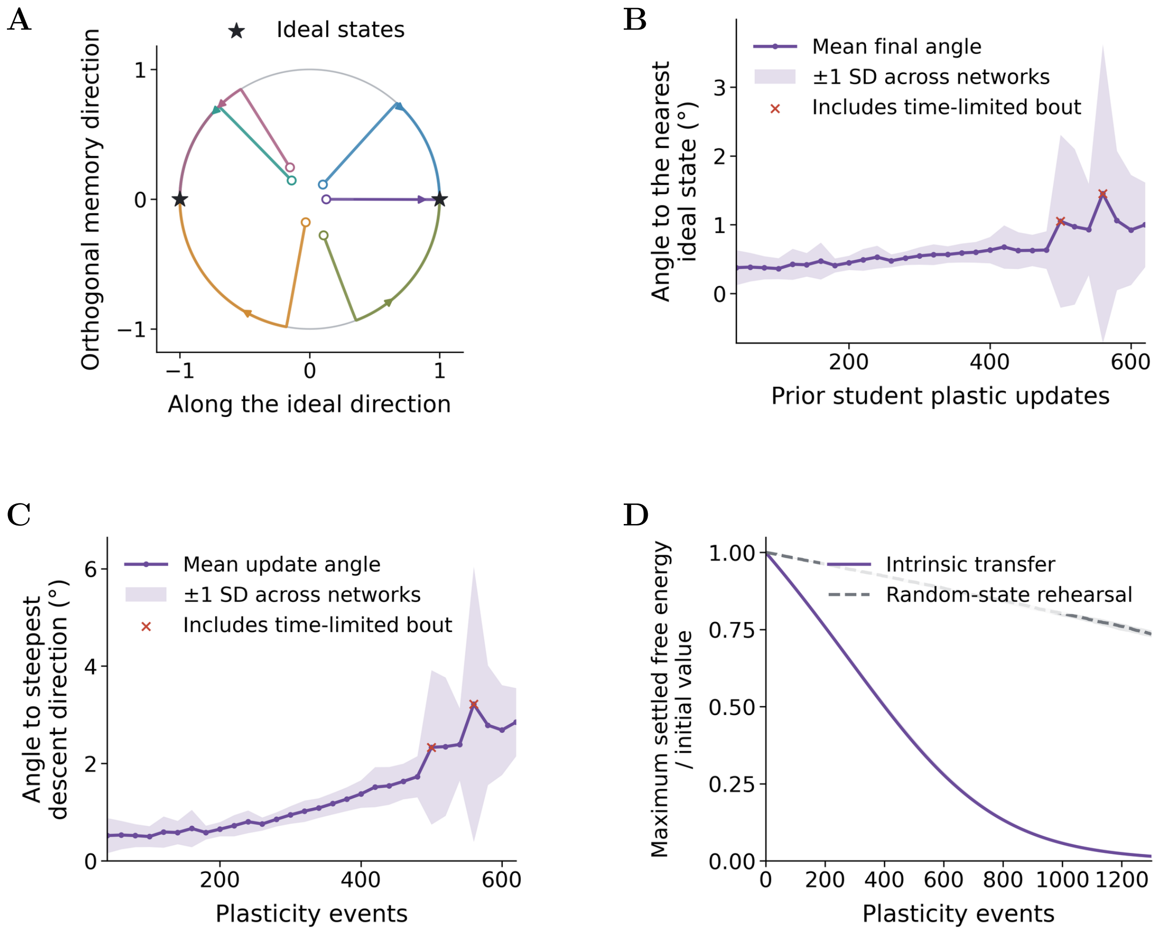

Fig 8. Coupled activity and local plasticity approximate minimization of the largest settled student free energy. Teacher weights were fixed, and student plasticity occurred only after activity relaxation. Ideal states and $\mathcal{F}_{\max}$ were calculated independently from the weights. **A,** Six teacher trajectories in a separate network with a two-dimensional memory space. The display projects the full unit teacher state onto this space without renormalizing it. The circle denotes unit memory states, stars the ideal states, open circles the initial states, small dots the endpoints, and arrows the direction of time. **B,** Final angle to the nearest ideal state after coupled relaxation, for students frozen at successive prior training checkpoints. Angles use the full activity vector and treat both signs of an ideal direction as equivalent. The curve averages ten initializations within each of 20 networks, then averages across networks; shading denotes one standard deviation across network means. **C,** One local plasticity event from each relaxed configuration in B, with $h = 10^{-3}$. The predicted and measured decreases in Eq. (23) are both divided by h; the dashed diagonal marks equality. **D,** Repeated plasticity events with $h = 0.005$, starting from students that store five of six teacher memories. Intrinsic transfer relaxes the coupled activity before each update; random-state rehearsal uses the student's equilibrium response to a fresh uniform unit teacher-memory state. Curves show medians across 20 networks and bands the interquartile range, after dividing each network's maximum by its initial value. Protocols are matched by event count. Network construction, settling criteria, and reference calculations are given in Materials and methods, Sections 6.6.1–6.6.3.

4 Discussion

However, CA1–PFC representational coordination was weaker and less faithful to the waking experience during sleep than during awake ripples. The authors interpreted sleep reactivation as more integrative and less like precise reinstatement. CLEAR LIMITATION DUE TO THE FACT THAT WE USES SIMPLE ASSOCIATIVE MEMORY NETWORKS

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. State explicitly that the analytic efficiency claim is local, that the exact $2/(1-\chi)$ factor assumes unconstrained weights and an already stored common component, and that faster complete transfer is supported numerically rather than by a global convergence or hitting-time theorem. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; effective cortical suppression of the teacher follows from maximizing recipient deficit; and the discrepancy-driven prioritization signal vanishes at complete transfer.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 State dynamics and diagnostics

[Teacher and recipient equations, activation, integration scheme, timestep, timescales, coupling, noise, normalization, settling, stopping, manifold leakage, and terminal occupancy.]

6.3 Prioritized, oracle, and random protocols

[Executable pseudocode or an algorithm box. Matching initializations, candidate states, one plasticity event per bout, thresholds, seed count, and uncertainty summaries.]

6.4 Outcome measures

[Define N_S, A_S, $\mathcal{D}_{\max}$, errors, occupancy, cumulative weight change, threshold events, oracle gap, nearest-memory distance, and cone confinement.]

6.5 Structured-memory transfer comparison

6.5.1 Controlled imbalanced corpus

The fast-transfer experiments used $P = 26$ unit-normalized memories arranged into three mutually orthogonal groups of sizes $n_g \in \{18, 5, 3\}$. Each group occupied a three-dimensional subspace, so the complete teacher memory space had rank $r = 9$. We assigned eight neurons to each independent direction, giving $d = 72$ teacher and student units. A fixed permutation of neuron coordinates removed any visible block structure from the recurrent weights without changing the memory geometry.

Within group g, memory p was constructed as

$$m_{g,p} = B_g c_{g,p}, \qquad c_{g,p} = \frac{c_0 + s_g\,(\cos\theta_{g,p}\,u + \sin\theta_{g,p}\,v)}{\sqrt{1+s_g^2}}, \qquad \theta_{g,p} = \frac{2\pi p}{n_g} + \phi_g, \tag{25}$$

where the columns of B_g form an orthonormal basis for group g, and c_0, u, v are orthonormal vectors in its three-dimensional coefficient space. The group phases ϕ_g were fixed across conditions. Evenly spaced residual directions satisfy a mean off-diagonal inner product of $-1/(n_g - 1)$, so choosing

$$s_g^2 = \frac{1-\rho}{\rho + 1/(n_g - 1)} \tag{26}$$

sets the mean cosine similarity between distinct memories in every group to ρ. The main comparison used $\rho = 0.70$. The correlation sweep used $\rho \in \{0, 0.35, 0.70\}$ while keeping the group bases, sizes, phases, rank, network parameters, and random seeds fixed. The $\rho = 0$ construction contains signed memories; all correlation-sweep simulations therefore used linear units.

The Gram matrix in Fig. 6(A) is $G = \mathcal{M}^\top \mathcal{M}$, where $\mathcal{M}$ contains the 26 memories as columns. Because every memory has unit norm, G_{pq} is their cosine similarity. The three diagonal blocks show within-group redundancy, whereas the zero-valued off-block entries follow from orthogonality between groups.

6.5.2 Matched transfer protocols

We constructed the frozen teacher weights from the same 26-memory corpus in both conditions. Both students started with $W_S = 0$ and used linear units, plasticity rate $\eta = 0.01$, integration interval $\Delta t = 0.05$, and one projected recurrent-weight update per plasticity event. The diagonal of W_S was reset to zero after every update. Each run contained 200,000 events. The state time constants were $\tau_S = 0.1$ and $\tau_T = 1$, the gains were $\pi_T = 1$, $\pi_S = 0.5$, $\pi_{TS} = 1$, and $\pi_{ST} = -0.5$ in the implementation sign convention. Teacher noise had amplitude 0.05, and teacher activity was normalized around unit norm.

In the random-presentation condition, an index was drawn uniformly from the 26 named memories at each event. The selected memory was clamped directly onto the student before applying its recurrent update. In the coupled condition, neither network was clamped. The teacher and student states advanced by one integration step under the coupled dynamics, after which the student received the same local recurrent update. Teacher weights remained fixed in both conditions. Model construction used seed 12, the random presentation schedule used seed 23, and stochastic state evolution used seed 12,357.

Uniform sampling over named memories weights the expected random-presentation update by the empirical second moment

$$C_m = \frac{1}{P}\sum_{p=1}^{P} m_p m_p^{\top}. \tag{27}$$

The 18-memory group therefore contributes more repeated updates than the 5- and 3-memory groups, although all three groups add the same number of independent directions. Stronger within-group correlation concentrates those repeated updates further onto the shared component of the large group.

6.5.3 Redistribution measures

For each named memory, we measured the recurrent reconstruction residual

$$r_p(t) = \|m_p - W_S(t) f(m_p)\|_2 . \tag{28}$$

The heatmaps in Fig. 6(C) show every $r_p(t)$ in group order. Under the linear activation used here, $f(m_p) = m_p$.

[Link with figure 6 B. and figure 7 :] Let $U_T = [u_1, \ldots, u_r] \in \mathbb{R}^{d\times r}$ be an orthonormal basis of the teacher memory space, with $r = 9$, and let $M_S(t) = I - W_S(t)$. We defined the subspace residual as

$$R_{\text{subspace}}(t) = \frac{\|M_S(t)U_T\|_F}{\sqrt{r}} = \left(\frac{1}{r}\sum_{j=1}^{r} \|M_S(t)u_j\|_2^2\right)^{1/2} . \tag{29}$$

reconstruct or represent ?

This basis-independent measure gives equal weight to every independent memory space direction. It equals one when $W_S = 0$ and vanishes exactly when the student reconstructs the complete teacher memory space.

We also measured the largest residual over any unit-norm direction in the teacher space,

$$R_{\text{worst}}(t) = \max_{\|a\|_2=1} \|M_S(t)U_T a\|_2 = \sqrt{\lambda_{\max}\big(U_T^{\top} M_S(t)^{\top} M_S(t) U_T\big)}. \tag{30}$$

This is the largest remaining hole in the student representation, including teacher-supported combinations that do not coincide with a named memory. Figure 6(B) reports both R_{subspace} and R_{worst}. Figure 7 reports only R_{subspace} so that the effect of correlation can be compared on matched axes.

6.6 Reference states and numerical tests of redistribution

The calculations below define the ideal teacher states and free-energy values used in Figure 8. The ideal teacher states after teacher convergence is the state that maximize free energy for the fixed recurrent weights student. The simulations used linear units, fixed teacher weights, and separate activity-relaxation and student-plasticity steps.

6.6.1 Calculating an ideal teacher state

The ideal teacher state depends on the teacher memory space and the current student weights. We calculate it while both weight matrices are fixed. Let $U_T \in \mathbb{R}^{d\times r}$ have orthonormal columns spanning $\mathcal{U}_T = \ker M_T$. These columns provide coordinates for the teacher memory space: every allowed teacher state can be written as $x_T = U_T a$, with $\|a\| = 1$. The basis was obtained from the right singular vectors of M_T with singular values below 10^{-9}.

The novelty matrix N_S, defined in Eq. (17), determines the student's settled free energy for any teacher state. To compare only states within the teacher memory space, we form

$$A_S = U_T^{\top} N_S U_T. \tag{31}$$

This smaller matrix describes how the student's remaining error varies among combinations of teacher memories. Equation (19) gives

$$q(U_T a; W_S) = \frac{\pi_{TS}}{2} a^\top A_S a, \qquad \mathcal{F}_{\max} = \frac{\pi_{TS}}{2} \lambda_{\max}(A_S). \tag{32}$$

To see why the largest eigenvalue gives the maximum, expand the unit vector a in an orthonormal eigenbasis of the symmetric matrix A_S. Its quadratic form is then a weighted average of the eigenvalues, with nonnegative weights summing to one. The largest possible value is the largest eigenvalue, attained by any unit vector in its eigenspace. Thus, for a leading unit eigenvector v_1, an ideal teacher state is

$$x_T^* = U_T v_1. \tag{33}$$

Its negative is equally valid. The corresponding student state is $x_S^*(x_T^*) = (I - N_S)x_T^*$. This calculation uses only the fixed weights and precisions; the simulated trajectory is not used to choose the reference state, and the reference state does not drive the simulation.

After each student-plasticity event, N_S and A_S are recalculated from the updated W_S, while U_T remains fixed. Matrix inverses in the analytical expressions were evaluated through Cholesky solves. Since N_S is positive semidefinite and has the same kernel as M_S, the maximum vanishes exactly when $M_S U_T = 0$, giving Eq. (24).

6.6.2 Distance to an ideal teacher state

Let V_1 contain an orthonormal basis of the leading eigenspace of A_S, and set $Q_* = U_T V_1$. For the full unit teacher state x_T, the nearest ideal state is

$$q_* = \frac{Q_* Q_*^\top x_T}{\|Q_*^\top x_T\|}, \qquad d_T = \|x_T - q_*\|, \qquad \theta_T = \frac{180}{\pi}\, 2\arcsin(d_T/2). \tag{34}$$

The angle θ_T is reported in degrees. If the projection is zero, all ideal states are orthogonal to x_T, and we assign $\theta_T = 90°$. For a simple leading eigenvalue, this calculation selects the nearer of the two signs of the ideal direction. Eigenvalues within $10^{-11} + 10^{-9}|\lambda_{\max}|$ of the maximum were treated as tied. Distances use the full activity vector, so they include departures from the memory manifold. A small angle implies proximity to an ideal state, but a large angle need not imply a large energy deficit when leading eigenvalues are close. For the empty student, $W_S = 0$, every unit teacher-memory state is ideal; the separate empty-student control therefore tests approach to the memory manifold rather than directional selection.

6.6.3 Simulation and plasticity protocol

Network construction and partial training. We generated 20 independent networks with $d = 48$ units per population and teacher-memory rank $r = 6$. Reduced QR decomposition of a Gaussian $d \times r$ matrix gave an orthonormal basis Q. The named unit memories were the columns of $P = QL^\top$, where $LL^\top = (1-\rho)I + \rho\mathbf{1}\mathbf{1}^\top$ and $\rho = 0.7$. Thus, distinct named memories had cosine similarity 0.7. With $C = QQ^\top$, the teacher weights were

$$(W_T)_{ij} = \frac{C_{ij}}{1 - C_{ii}} \quad (i \neq j), \qquad (W_T)_{ii} = 0. \tag{35}$$

This construction represents the span of P exactly. Teacher weights remained fixed and all student updates preserved zero diagonals.

For Fig. 8B,C, students started at $W_S = 0$. Each prior training event sampled a named memory uniformly, fixed the teacher at that memory, set the student to its exact equilibrium response, and applied Eq. (22) with $h = 0.25$. We froze student weights after 40 to 320 updates in increments of 20. Ten full-sphere teacher initializations per network were reused at every checkpoint, with student activity initialized to zero. Each checkpoint in B averages the ten final angles within each network, then reports the mean and sample standard deviation of the 20 network means. The separate rank-two illustration in A used 160 prior training events and six initializations.

Coupled relaxation. We used $\pi_T = 1$, $\pi_S = 0.5$, $\pi_{TS} = 1$, $\tau_T = 10$, and $\tau_S = 1$. Writing $S_T = M_T^\top M_T$, we set $\kappa = 0.45\pi_T\sigma_{\min}^2$, where $\sigma_{\min}$ is the smallest nonzero singular value of M_T. The integrated dynamics were

$$v_T = \frac{-\pi_T S_T x_T + \kappa(x_T - x_S)}{\tau_T}, \qquad \dot{x}_T = v_T - x_T \frac{x_T^\top v_T}{x_T^\top x_T}, \qquad \dot{x}_S = \frac{\pi_{TS} x_T - H_S x_S}{\tau_S}. \tag{36}$$

The teacher derivative preserves norm without projecting activity into its memory space. There was no noise or plasticity during relaxation. Independent bouts began with a normalized isotropic Gaussian teacher state and zero student activity.

Integration used float64 LSODA, relative tolerance 2×10^{-7}, absolute tolerance 2×10^{-9}, and maximum step 500. Integration blocks began at 50 time units and doubled to a maximum block length of 20,000. At 11 sampled times over the final 50 time units of each block, both $\|\dot{x}_T\| < 10^{-7}$ and $\|x_S - x_S^*(x_T)\| < 10^{-6}$ had to hold before stopping. Teacher norm was restored after each block, and each bout was capped at 600,000 time units. All 3,200 selection bouts, including the empty-student controls, and all 14,000 intrinsic-cycle bouts met these criteria. No networks or initializations were excluded.

Single-event prediction. At student equilibrium and an ideal teacher state with a simple leading eigenvalue, differentiation of the settled maximum gives

$$P_0 \nabla_{W_S} \mathcal{F}_{\max} = -\pi_S P_0 \big[(M_S x_S^*) x_S^{*\top} \big]. \tag{37}$$

The derivatives through the minimizing student state and maximizing teacher state vanish by stationarity. The zero-diagonal projection restricts the gradient to admissible weights. Consequently, setting $G_W = \pi_S P_0[(M_S x_S) x_S^\top]$, an ideal event satisfies

$$\frac{\mathcal{F}_{\max}(W_S) - \mathcal{F}_{\max}(W_S + hG_W)}{h} = \|G_W\|_F^2 + O(h).$$

For each of the 3,000 partially trained selection bouts, we used the observed terminal student state to form G_W, applied one event with $h = 10^{-3}$, and independently recomputed the maximum. These are separate tests from identical frozen checkpoints, not successive events in Fig. 8D. Companion step-size checks repeated each event from the same pre-event weights and states at $h \in \{0.003, 0.001, 0.0003\}$. Relative prediction errors excluded only predictions below 10^{-12}; all events were retained in the main comparison.

Repeated cycles and random-state reference. For Fig. 8D, both protocols began with the same student predictor constructed by Eq. (35), using the projector onto the span of the first five named memories. Each protocol applied 700 events with $h = 0.005$. Intrinsic transfer relaxed the coupled activity before every update. Its first bout began from a random full-sphere teacher state and zero student state; subsequent bouts carried forward the preceding terminal activities. Random-state rehearsal drew a fresh uniform unit vector $a \in \mathbb{R}^6$, set $x_T = U_T a$, and used the exact equilibrium response $x_S^*(x_T)$ for the same local update. The maximum was recomputed after every event and divided by that network's initial maximum before calculating the median and 25th–75th percentiles across networks. The protocols were matched by plasticity-event count, not relaxation time.

Random numbers used NumPy `default_rng`, with network seeds $20260905 + n$ for $n = 0, \ldots, 19$. Offsets of 100,000, 200,000, and 300,000 set the training, selection-initialization, and repeated-cycle streams. The illustration used seeds $20260905 + 900000$ for the network and $20260905 + 900001$ for the starts. The workbench checked the settled-energy identity, the weight gradient by finite differences, zero diagonals, and agreement with the repository dynamics. Selected trajectories were also checked with tenfold tighter solver tolerances and half the maximum integration step.

Supporting information

- **S1 Appendix.** Full proofs of Proposition ??, Theorem ??, Corollary ??, and Corollary ??; projected power iteration; envelope steps; antipodal maximizers; feasible projected descent; and the local correlated-memory rate comparison.

- **S1 Fig.** Ascent, zero-drive, and descent teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.

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